MIL-C-8412B (USAF)
18 August 1969

SUPERSEDING
MIL-C-8412A (USAF)
6 May 1957

MILITARY SPECIFICATION

COMPASS, MAGNETIC, SLAVED GYRO, TYPE J-4

1. SCOPE

1.1 This specification covers one type of an airborne slaved gyro magnetic compass, designated as Type J-4.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

L-P-378	Plastic Film, Polyethylene Thin Gage
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard
PPP-C-843	Cushioning, Material, Cellulosic
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock # Container Grade # and Cut Shapes

Military

MIL-E-1E	Electron Tube, General Specification For
MIL-P-116	Preservation, Methods of
MIL-C-5015	Connector, Electric, "AN" Type
MIL-I-5126	Indicator, Directional, Gyroscopic, Magnetic Compass, Type V-7A
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5541	Chemicals Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-S-7742	Screw Thread, Standard, Optimum Selected Series, General Specification For

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

MIL-C-7769	Cushioning Material, Uncompressed Round Fiber for Packaging
MIL-P-7788	Plate, "Plastic", Lighting
MIL-A-8625	Anodic "Coatings", For Aluminum and Aluminum Alloys
MIL-D-1000	Drawing, Engineering and Associated List
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

STANDARDS

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standard Order of Precedence For the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-681	Identification Coding and Application of Hookup and Lead Wire
MIL-STD-838	Lubrication of Military Equipment
MS33558	Numerical and Letter, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

*3.1 Preproduction. This specification makes provisions for First Article Inspection.

3.2 Components. The Compass, Magnetic, Slaved Gyro, Type J-4, shall consist of:

ITEM NO.	QUANTITY	DESCRIPTION	REFERENCE PARAGRAPH
1	1	Directional Control	3.7.1
2	1	Servo-Amplifier	3.7.2
3	1	Control Panel	3.7.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

*3.2.1 Associated Equipment. Unless otherwise specified, the following shall be used as supplementary equipment with the Type J-4 compass:

QUANTITY	DESCRIPTION	SPECIFICATION
1	Gyroscope, Rate CN-436/AJN-3	MIL-C-25773
1	Indicator, Directional Gyro, Magnetic Compass, Type V-7A	MIL-I-5126
1	Detector, Magnetic, Azimuth DT-309/AJN	MIL-D-25521
1	Transmitter; Remote Compass Type C-2 (Unstabilized)	MIL-T-25193

3.3 Selection of Specifications and Standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard Parts. With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials

3.4.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungistatic and corrosion protective treatment, anodizing of the aluminum alloy parts, and cadmium plating of steel parts are not applicable for parts within hermetically sealed enclosures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

3.4.2 Nonmagnetic Materials. Nonmagnetic materials shall be used for all parts of the compass, except where magnetic materials are essential.

3.4.3 Fungus-Proof Materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrient for fungi must be used, such materials shall be treated with a fungistatic agent, as approved by the procuring activity.

3.4.4 Protective Treatment. When materials are used in the construction of the compass that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes of atmospheric conditions shall be avoided.

3.5 Design and Construction. The Type J-4 compass shall be designed to operate as a directional reference, either as a directional gyro or by reference to the magnetic meridian. The compass shall be designed to synchronize internally as a system independent of unique orientation of the direction of the spin axis of the gyroscope about its vertical axis.

3.5.1 General

3.5.1.1 Maintenance. The compass shall be designed so that overhaul, including replacement of worn parts, adjustment, balance and calibration may be accomplished by an instrument repairman provided with appropriate tools and equipment and trained in the repair of gyroscopic and related equipment.

3.5.1.2 The compass shall be designed for installation and operation in fighter, short range bomber and cargo type aircraft. The compass shall be suitable for full maneuverability without damage during high rates or rotation not to exceed 50 rpm maximum.

3.5.1.3 The compass shall be constructed so that no parts will work loose in service. The compass shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation, and service use.

3.5.1.4 The components of the Type J-4 compass shall be protected to insure that sand, dust, humidity and fungus producing conditions in service life will not interfere with the proper operation of the overall compass.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

3.5.1.5 The components of the Type J-4 compass shall be suitable for continuous operation for extended periods of time under all possible service conditions.

3.5.1.6 Mounts. The mounts provided for the various components of the compass shall be designed so that the compass is not affected by vibration and shall provide adequate protection to insure maximum service life. The design shall allow for ease of installation and service. The mounts shall be subjected to a vibration test, procedure V of MIL-E-5272.

3.5.2 Power Requirements.

*3.5.2.1 The compass shall operate on 100 volt-amperes AC, 3-phase, 400 hz AC, power and 28 watts d.c. and the power demand shall not exceed 75 volt-amperes AC and 15 watts d.c.

3.5.2.2 The compass shall start and operate without excessive power requirements while at an ambient temperature of -54°C (-65°F).

3.5.3 Power Connectors

3.5.3.1 Unless otherwise specified, the power connectors shall be in accordance with MIL-C-5015 or of a type acceptable to the procuring activity.

3.5.3.2 Power connectors shall be protected when not in use by a dust-tight cover acceptable to the procuring activity.

3.5.4 Environmental Conditions. The compass and its individual components shall function properly under each of the following conditions:

a. Temperature: The compass shall be designed to operate throughout a temperature range of -54°C to 71°C (-65°F to 160°F). Continued exposure to temperatures as low as -85°C (-85°F) shall not affect the operation of the compass over the temperature range indicated.

b. Humidity: A relative humidity of 95 ±5 percent at a temperature of 71°C ±2°C (160°F ±3°F).

c. Altitude: A pressure altitude range from 0 to 70,000 feet.

d. Salt Spray: Exposure to atmosphere containing salt laden moisture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

e. Vibration: Conditions of vibration incident to normal use in aircraft or any variation in frequency of vibration between the limits of 300 and 3000 hz per minute and variations in amplitude in any plane from 0.002 to 0.020 inch, or any combination thereof.

f. Sand and Dust: Exposure to airborne sand and dust particles encountered in normal and desert operation.

3.5.5 Service Life: The equipment shall have a mimimum service life of 5000 hours with failures limited as specified in 3.5.6. Parts requiring replacement within this interval and the excepted life of such parts shall be specified by the contractor and be subject to the approval of the procuring activity.

3.5.6 Reliability. Reliability of the equipment shall be a major design criterion. The design and construction of the equipment shall be such as to provide maximum practical periods of time between failures of equipment. Failure shall be defined as any malfunction which results in degraded performance regardless of the cause (careless handling or involuntary destruction expected). The Type J-4 compass system shall have a mean-time-to-failure of at least 500 hours.

3.6 Performance. The performance of the compass shall be such that the results on the components or over-all compass shall be within the tolerances specified for the tests described in section 4 of this specification.

3.7 Details of Components

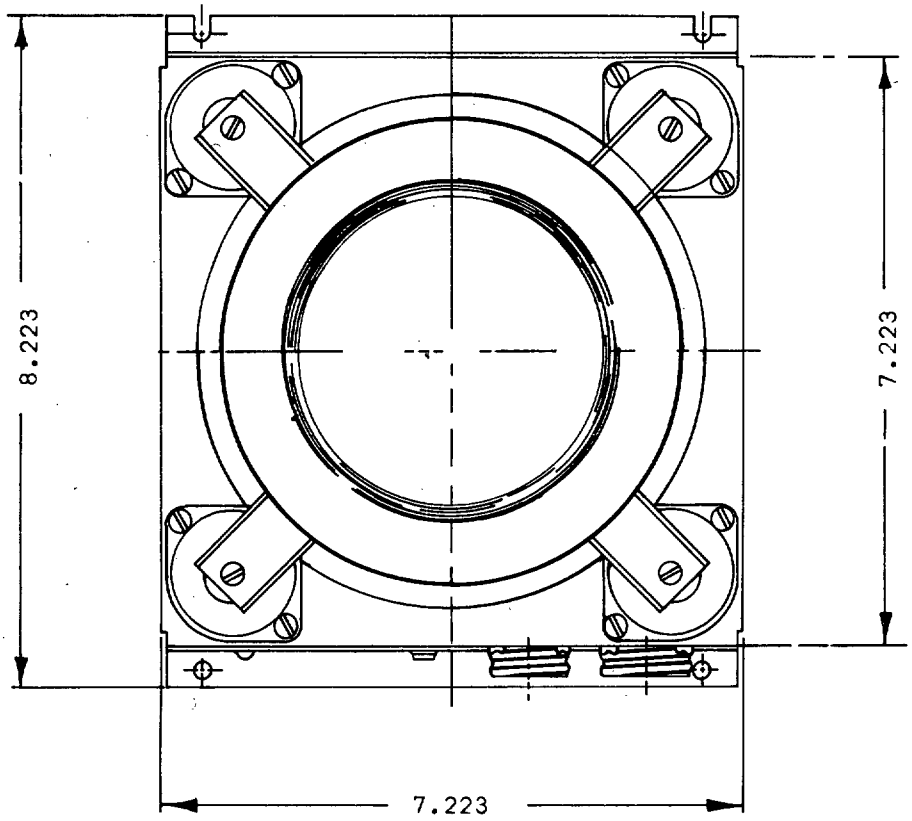
3.7.1 Directional Control. The gyro element of the directional control shall be symmetrical about a plane normal to the spin axis at its midpoint and shall have no appreciable balance shift due to temperature changes. The gyro shall be nonpendulous and balanced about both axes to minimize acceleration effects. The directional control shall feed an azimuth servo in which slewing and fast synchronization is accomplished.

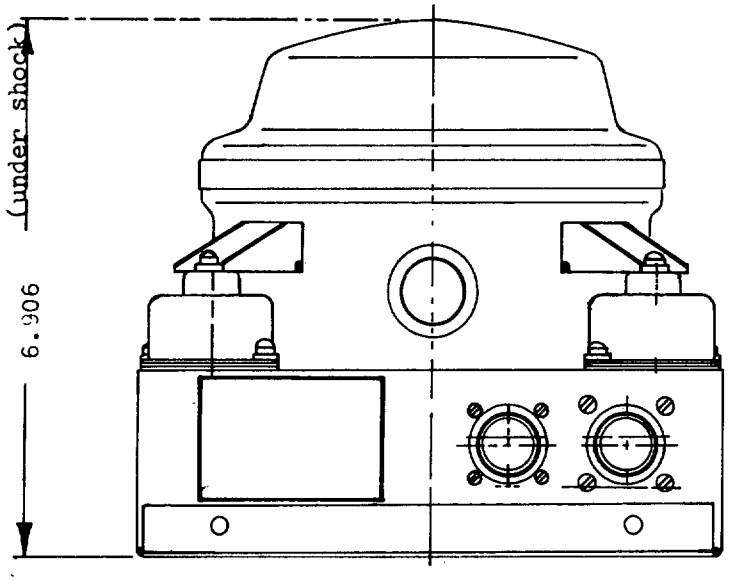
3.7.1.1 General. The configuration of the directional control shall conform to figure 1.

3.7.1.1.1 Hermetic Seal. The directional control shall be hermetically sealed to make it impervious to environmental conditions.

3.7.1.1.2 Window. A window shall be provided in the case above the mount through which the leveling axis may be viewed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.





DIMENSIONS ARE IN INCHES.

CONFIGURATIONS AS SHOWN ARE FOR MAXIMUM DIMENSIONS ONLY.

DIRECTIONAL CONTROL
FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

3.7.1.1.3 Rotor Enclosure. The gyro rotor shall be suitably enclosed to minimize drift due to windage.

3.7.1.1.4 Stops. Stops on the inner axis shall be provided in positions such as to minimize reversal errors.

3.7.1.1.5 Mounting. The directional control shall be provided with a mounting base and shall be supported in the horizontal plane containing the center of gravity of the instrument by adequate vibration isolators of a type acceptable to the procuring activity. The mounts shall exhibit rigidity about the azimuth axis to the extent that there shall be no rotation about the azimuth axis in excess of 0.02° as a result of any vibration between 0 and 18 hz, with a maximum amplitude of 0.20 inch. The resonant period of the mounts shall be between 5 and 10 hz in the horizontal and vertical modes. The torsional resonant period shall be greater than 18 hz. The transmissibility of the vibration isolation system shall not exceed a ratio of 5.5:1.

3.7.1.2 Power Requirements. The directional control shall be electrically operated on 115V, 3-phase, 400 hz ac with phase rotation of A, B, and C.

3.7.1.3 Transmitting Synchros. Two synchros shall be attached to the azimuth axis of the gyro to transmit signals proportional to the position of the gyro. The synchros shall be of the three-phase wye connected stator and single phase rotor types.

3.7.1.3.1 Azimuth Transmitter. The electrical characteristics of the azimuth transmitter shall be such as to operate the azimuth servo channel in the servo-amplifier. The transmitter shall have an electrical error spread of 12 minutes or less and an accuracy of 8 minutes. No deterioration in the operation of the azimuth servo channel shall occur when a control transformer, having the following characteristics and loaded with a minimum impedance of 10,000 ohms, is paralleled with the control transformer in the azimuth servo channel:

Input Impedance: 80 +j400 ohms
Rotor Resistance: 42 ohms
Stator Voltage, Line-to-Line: 11.8 volts maximum

3.7.1.3.2 Autopilot Transmitter. The autopilot transmitter shall operate from a rotor excitation of 115V, 400 hz, and the maximum stator output voltage, line-to-line, shall be 18.2 volts open circuit. The transmitter shall have an electrical error spread of 15 minutes or less. The trans-

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

mitter shall have 5 separate leads terminating in separate pins of the electrical connector. The transmitter leads shall be shielded.

3.7.1.4 Slaving. The gyro shall be designed so that it can be slaved to an azimuth signal. The normal slaving rate shall be 1° to 2° per minute, exclusive of earth's rate, and shall provide constant precession rates throughout a wide range of tilt angle.

3.7.1.5 Random Wander. When the gyro is mounted in its normal operating position and tested on a 0.5° and 7.5° Scorsby table or 0.5° and 7.5° equivalent roll, pitch and yaw mechanism, the random wander of the gyro for any position of the vertical gimbal shall not be more than ± 3° per hour from the computed apparent drift for the latitude of operation.

3.7.1.6 Balance. The gyro shall be so balanced that the apparent drift on any heading is equal to the product of the sine of the latitude of operation and 15° per hour in a counterclockwise direction in the Northern Hemisphere and in a clockwise direction in the Southern Hemisphere, as indicated on a Type V-7A repeater indicator (MIL-I-5126, or other means approved by the procuring activity) connected to the autopilot transmitter.

3.7.1.7 Vertical Gimbal. The directional control shall be so designed that there shall be no excessive spinning or oscillation of the vertical gimbal when power is connected to, or removed from, the compass.

3.7.1.8 Leveling Device. The gyro shall be leveled at a rate of 2° to 4° per minute exclusive of earth's rate. The type of leveling pickoff utilized in this design shall have minimum frictional, vibration and acceleration errors.

3.7.1.9 Weight. The weight of the complete directional control, including the mount, shall not exceed 13 pounds.

3.7.2 Servo-Amplifier. The servo-amplifier shall fulfill the necessary servo and power requirements for proper compass operation. It shall also contain the azimuth servo integral therein. The configuration of the servo-amplifier shall conform to figure 2.

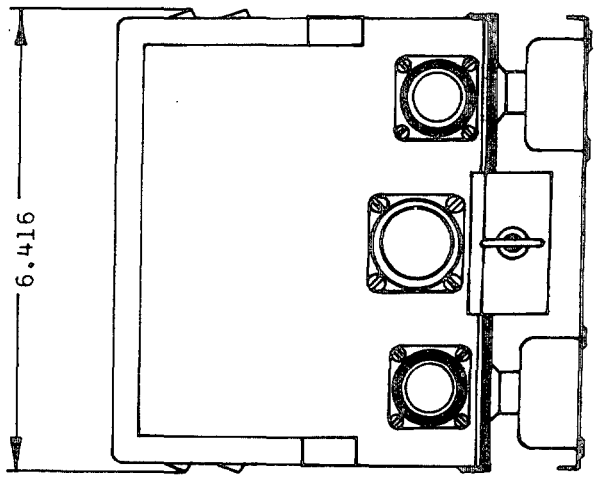
3.7.2.1 Power Requirements. The amplifier shall operate from 115V, single phase, 400 hz ac and 27.5v dc.

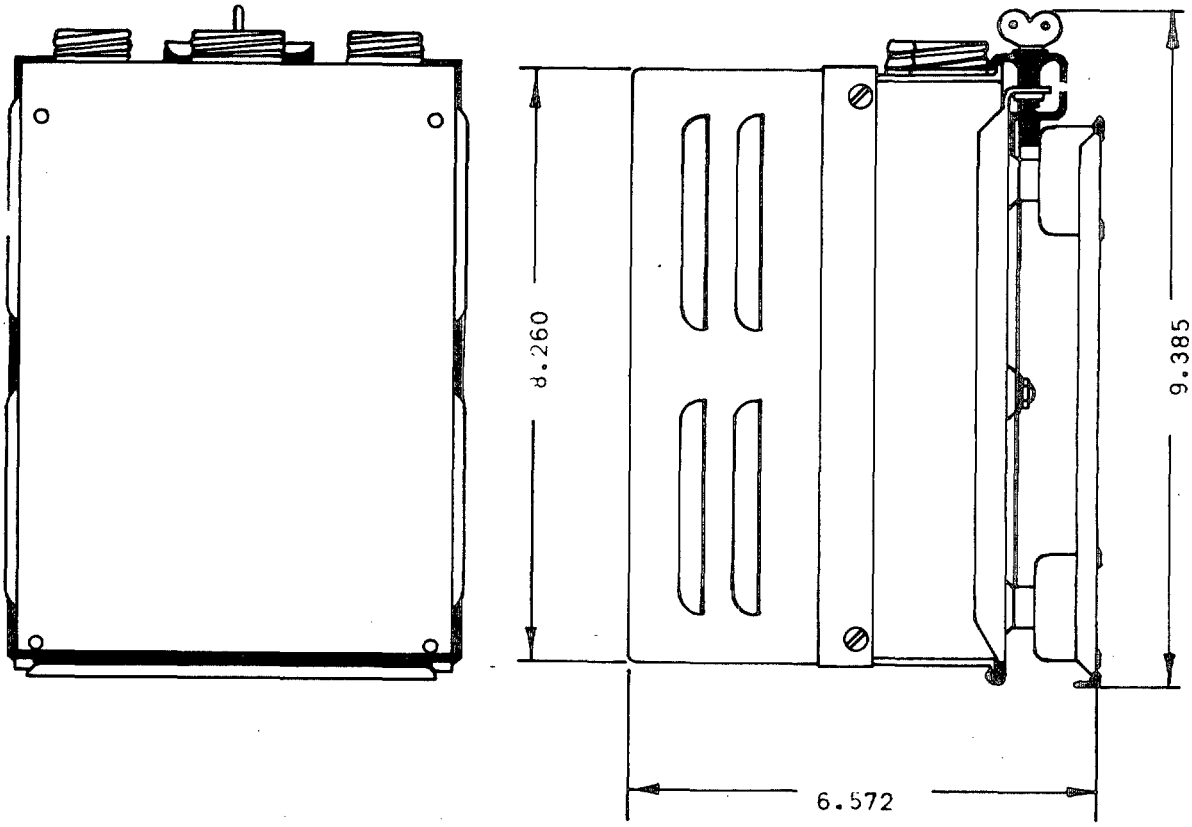
3.7.2.2 Data Transmitters. Two data transmitters whose rotors are positioned by the azimuth servo channel shall be provided for the electrical transmission of heading reference to auxiliary equipment. All connections to rotors and stators shall be brought out to separate pins in the electrical connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

DIMENSIONS ARE IN INCHES.

CONFIGURATION AS SHOWN ARE FOR MAXIMUM DIMENSIONS ONLY.





SERVO AMPLIFIER
FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

3.7.2.2.1 Indicator Transmitter. A four-power transmitter with an electrical accuracy of 10 minutes or less shall be provided for the operation of remote indicators such as the Type V-7A. The rotor excitation shall be 115V, 400 hz and the maximum output voltage, line-to-line, shall be 18.2 volts open circuit.

3.7.2.2.2 Precise Data Transmitter. A transmitter having an electrical accuracy of 7 minutes or less shall be provided for the operation of precise control synchros, Pioneer AY200 series, or equal. It shall have the following characteristics:

Input Voltage: 26V, 400 hz
Input Impedence: 25"+j115"ohms
Output Voltage (Maximum Line-to-Line): 11.8 volts

The transmitter shall be capable of operating two ID-249 ( )/ARN, ID-351A/ARN, ID-357/ARN or ID-387/ARN, two ID-250A/ARN and two ID-251/ARN course indicators simultaneously.

3.7.2.3 Amplifier and Servo Channels.

*3.7.2.3.1 The amplifier channels shall be suitable for continuous operation for extended periods of time under all possible service conditions and shall utilize tubes in accordance with MIL-E-1E.

3.7.2.3.2 The servo channels shall so control the servo motors that the response of the compass is in accordance with the conditions specified herein.

3.7.2.4 Azimuth Servo Channel. The servo-amplifier shall contain a fast servo system with a follow-up rate of at least 7 rpm to maintain synchronization with gyro movement. For velocities of 5 rpm or less the error shall not be greater than 0.2° per rpm. The undamped natural frequency shall be 6 hz with damping approximately 60 percent of critical.

*3.7.2.5 Slaving Channel. The slaving channel shall incorporate a control transformer which receives a signal from a DT-309/AJN detector, or electrical equivalent, and whose rotor is positioned by the azimuth servo channel. The output of the slaving channel amplifier shall slave the directional control at the rates given in 3.7.1.4.

3.7.2.6 Slaving and Leveling Cutout. Provisions shall be made for leveling cutout and for deslaving the gyro. This shall be accomplished by applying 28V dc to two of the pins of the electrical power connector. Leveling cutout shall operate under all modes of operation. Slaving cutout shall be effective when the compass is in the magnetic mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

*3.7.2.7 Automatic Magnetic Synchronization. The azimuth servo channel shall be automatically synchronized with the DT-309/AJN detector, or electrical equivalent, upon the initial application of power to the compass, when the control panel function selector switch is in the MAG Position. The compass shall also be automatically synchronized when it is switched from the GD mode of operation to the MAG mode of operation. Automatic synchronization shall be accomplished at a rate of approximately 7 rpm.

3.7.2.7.1 After the automatic synchronization cycle has been completed or when the compass is in the DG mode of operation, the syncbronization motor shall be actuated by the control panel synchronizer control to accomplish manual synchronization or to position the data transmitters to any desired reference.

3.7.2.8 Weight. The weight of the complete servo-amplifier, including the mount, shall not exceed 11 pounds.

3.7.2.9 Servo-Amplifier (For Use in F-104 Aircraft Only). This repackaged servo-amplifier for use in only the F-104 aircraft shall meet servo-amplifier requirements specified in 3.7.2 through 3.7.2.7 inclusive. The configuration of this servo-amplifier shall conform to figure 3.

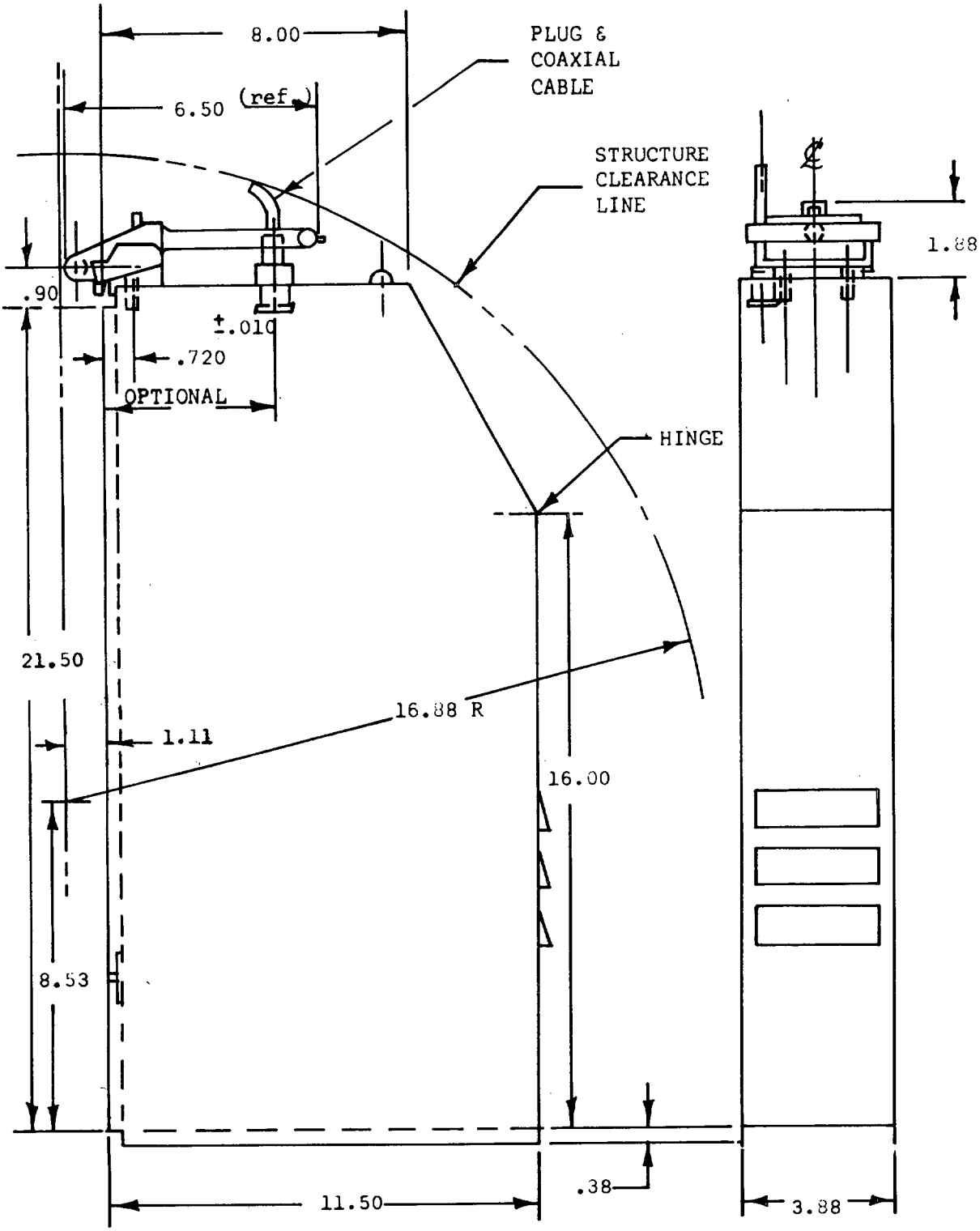
3.7.2.9.1 Weight. The weight of the complete repackaged servo-amplifier shall not exceed 15 pounds.

3.7.3 Control Panel. A control panel, designed in accordance with the requirements of MIL-C-6781, MIL-P-7788 and MIL-F-25173, shall provide the controls and indications necessary to operate the compass. The dimensions and general layout of the control panel shall conform to figure 4.

*3.7.3.1 Function Selector Switch. A switch shall be provided to select the mode of operation of the compass. When the switch is in the MAG position, the compass shall operate as a magnetic compass and synchronize with the DT-309/AJN detector, or electrical equivalent. When the switch is in the DG position, the compass shall operate as a latitude corrected directional gyro.

3.7.3.2 Latitude Correction Controller. A manually set controller shall be provided to correct the compass (when in the DG mode of operation for the apparent drift of the gyro due to earth's rotation). The correction shall be at a rate equal to the product of the sine of the indicated latitude and 15 per hour, within a tolerance of $\pm 1^{\circ}$ per hour. The correction shall be clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere as selected by a hemisphere switch. The hemisphere switch shall be controlled from a shaft with a screwdriver slot for turning. The latitude dial shall be marked with latitudes from 0° to 90° and graduated in 2-degree increments with major divisions and numerals every 10°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

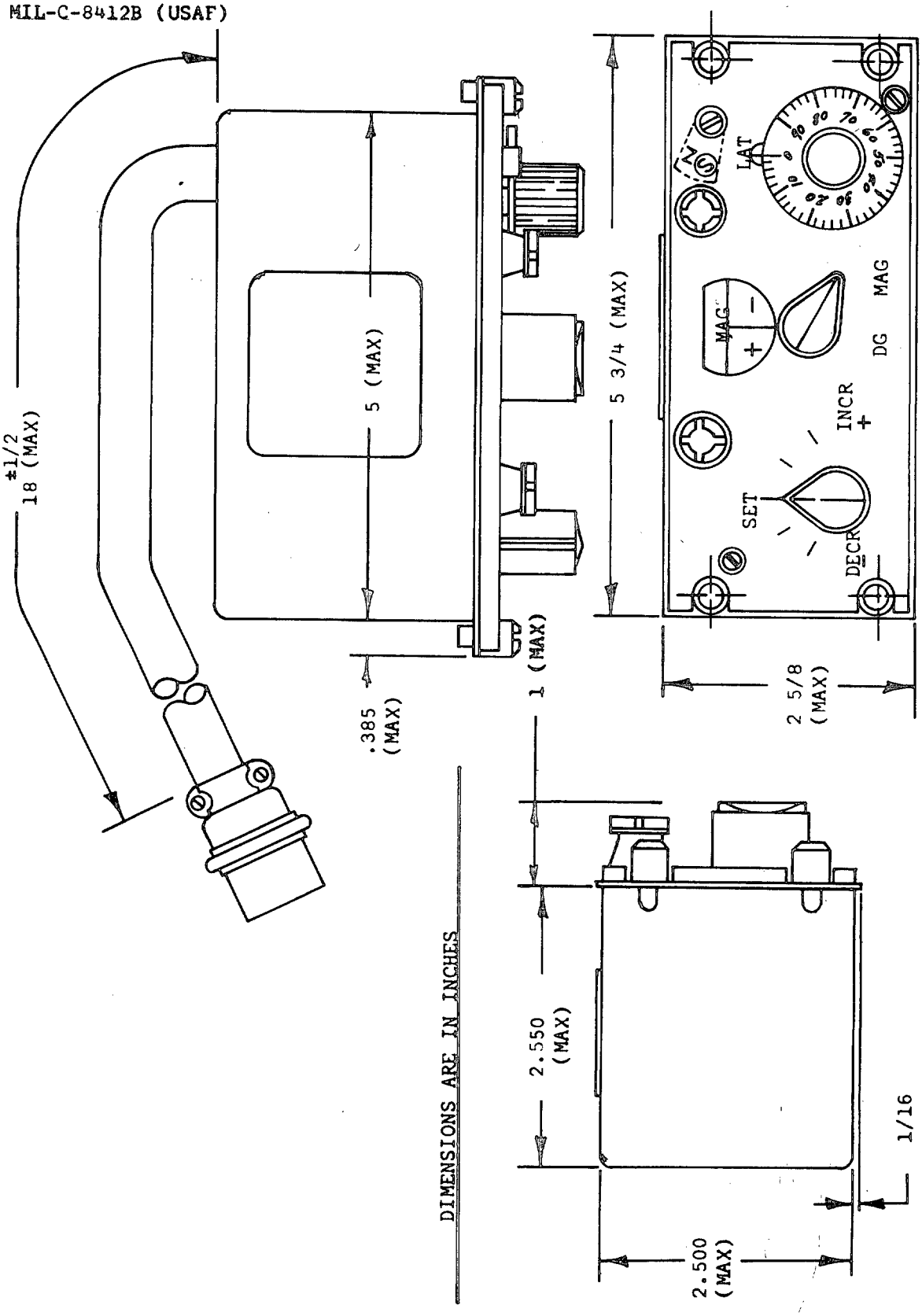


DIMENSIONS ARE IN INCHES
TOLERANCES ±.03 EXCEPT WHERE NOTED

SERVO-AMPLIFIER (FOR F-104 ONLY)

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.



COMPASS CONTROL PANEL
FIGURE 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

*3.7.3.3 Annunciator. An annunciator, other than signal lights, shall be provided to indicate synchronization to within ±0.25° between the compass and the DT-309/AJN detector, or electrical equivalent, when in the magnetic mode of operation. When the compass is not synchronized, the annunciator shall indicate in which direction the synchronizer must be turned to avoid an ambiguity. The annunciator lighting and markings shall afford ease of viewing under all ambient lighting conditions.

3.7.3.4 Synchronizer. A manual control shall be provided to effect synchronization of the data transmitters when the compass is operating as a magnetic compass and to position the data transmitters to any desired reference when the compass is operating as a directional gyro. The control shall be of the spring return-to-center type providing slewing rates of approximately 0.5 rpm and approximately 7 rpm in both the clockwise and counterclockwise direction. A Type V-7A indicator, or other means acceptable to the procuring activity, connected to the indicator data transmitter of 3.7.2.2.1 shall rotate in a clockwise direction when the synchronization control is rotated in a clockwise direction. The synchronization control marking shall correspond to the annunciator markings and shall indicate in which direction the control is to be turned to synchronize the compass.

3.7.3.4.1 Autopilot Decoupling. An additional set of contacts shall be mounted to the manual control to automatically decouple the autopilot output during slewing operation. These contacts shall normally be closed. They shall be opened when slewing is being accomplished.

3.7.3.5 Markings. All markings shall be durable to withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MS33558.

3.7.3.5.1 Numerals. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to the direct and distinct relation of a graduation to a numeral shall be cause for rejection.

3.7.3.5.2 The dimensions of the markings shall be as follows:

	Height or Length (inch ±0.010)	Width of Line or Graduation (inch ±0.005)
Type J-4 Compass	0.093	MS33558
All Other Letters and Words	0.125	MS33558
Numerals	0.125	MS33558
2° Graduations	0.093	0.010
10° Graduations	0.156	0.020
"+" and "-"	0.125	0.020

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

The width of the 2° graduations shall be on the minimum side of the tolerance wherever possible.

3.7.3.5.3 Other Markings. All markings not otherwise specified shall be finished in durable dull black.

3.7.3.6 Weight. The weight of the complete control panel shall not exceed 1 pound, 10 ounces.

*3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.9 Physical Properties.

3.9.1 Pivots, bearings and gears shall neither bind nor shake in operation. These parts and the parts necessary for electrical commutation (i.e., slip rings, brushes and rotary contacts) shall be as near frictionless as possible, except in any case where friction is required.

3.9.2 Relays. Relays, if used, shall be of the type acceptable to the procuring activity.

3.9.3 Screw Threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

*3.9.4 Lubrication. Where required, lubrication shall be accomplished in accordance with MIL-STD-838.

*3.9.5 Soldering. Soldering shall be in accordance with MIL-STD-454.

*3.9.6 Color Coding. All wiring shall be suitably color coded in accordance with MIL-STD-681 where applicable.

3.9.7 Finishes.

3.9.7.1 Aluminum Alloy Parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Dials, small holes and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.9.7.2 Steel Parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416.

*3.10 Radio Interference. The compass shall not cause any radio noise above the levels specified when tested in accordance with MIL-STD-461, except as specified in 4.6.32.

3.11 Weight. The total weight of the Type J-4 compass shall not exceed 25-1/2 pounds.

3.12 Identification of Product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Workmanship.

3.13.1 General. The compass, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.13.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the compass, they shall be held or limited accordingly.

3.13.3 Screw Assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt, or threads.

*3.13.4 Cleaning. The compass shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign material before final assembly. Resin flash that may crumble, shall be removed.

3.13.5 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

3.13.6 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight and loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

4. Quality Assurance Provisions.

*4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

1. First Article Inspection (See 4.3)
2. Quality Conformance Inspection (See 4.4)

*4.3 First Article Inspection.

4.3.1 First Article Inspection Samples. The contractor shall conduct inspections on three preproduction sample equipments to determine that the design of the equipments meets the requirements specified herein.

4.3.2 First Article Inspections. First Article inspections shall consist of all tests described under 4.6, subject to the test conditions specified herein.

*4.4 Quality Conformance Inspection.

4.4.1 Quality Conformance Inspection shall consist of:

a. Individual tests.

b. Sampling Plan and tests

4.4.2 Individual Tests. Each compass (unless otherwise specified that the component may be tested individually) shall be subjected to the following tests as described under 4.4 of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

TESTS	APPLICABLE PARAGRAPH
Examination of Product	4.6.1
Operating Speed	4.6.2
Azimuth Servo Channel Operation	4.6.3
Follow-Up Rate	4.6.4
Normal Slaving Rate	4.6.5
Random Wander	4.6.7
Latitude Correction	4.6.9
Balance	4.6.10
Gyro Card Stability	4.6.11
Gimbal Freedom	4.6.12
Spinning and Oscillation	4.6.13
Leveling	4.6.14
Polarity	4.6.15
Annunciator	4.6.16
Static Settling Test	4.6.17
Dynamic Accuracy	4.6.18
Scale Error	4.6.19
Turn Cutout Magnetic Slaved Operation	4.6.20
Synchronization	4.6.21
Autopilot Slewing Decoupling	4.6.22
Reliability Test	4.6.23

4.4.3 Sampling Plan and Tests. Unless otherwise specified, five compasses shall be selected at random from each quantity of 500 or fraction thereof manufactured on the production run and subjected to the following tests as described under 4.6.

TESTS	APPLICABLE PARAGRAPH
Random Wander	4.6.7.1
Dielectric Strength and Insulation Resistance	4.6.24
Low Temperature Operation	4.6.25
High Temperature Operation	4.6.26
Power Consumption	4.6.27
Voltage and Frequency Variation	4.6.28
Low Field Strength	4.6.29
Rotational Test	4.6.30
Vibration	4.6.31
Radio Noise Interference	4.6.32

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.4.4 Rejection and Retest. When one item selected from a production run fails to meet this specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.4.5 Individual Tests May Continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

4.4.6 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Testing Requirements.

4.5.1 Test Conditions.

4.5.2 Standard Atmospheric Conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it shall be understood that the test will be made at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25°C (77°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.3 Attitude. Unless otherwise specified, the compass shall be tested in its normal operating position.

4.5.4 Vibration. A vibration stand shall be used which will meet the conditions stipulated under procedure V of MIL-E-5272.

4.5.5 Voltages. Unless otherwise specified, all tests shall be conducted with the compass operating on 115V ±5V, 400 ±10hz, 3-phase ac and 27.5V ±1VDC.

4.5.6 An electrical equivalent may be substituted for the magnetic detector for all but preproduction tests. Unless otherwise specified, all tests shall be conducted in a field of approximately 180 milligauss.

4.5.7 Unless otherwise specified, a Type V-7A indicator in proper operating condition or other means acceptable to the procuring activity shall be connected to the indicator transmitter of 3.7.2.2.1 to provide an indicator for the compass. Where the tolerance specified for a particular test requires

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

an accuracy greater than that specified for the Type V-7A indicator, a precise indicator accurate to within 0.1° shall be used.

4.5.8 Test Procedures. If the complete procedure and method for conducting tests are not stated in 4.4, they shall be prepared by the contractor and submitted to the procuring activity for approval at least 30 days prior to tests. The approved test procedures shall be considered a part of this specification. The right is reserved by the procuring activity to modify the tests or require any additional tests considered necessary to determine compliance with the requirements of this specification.

4.6 Test Methods.

4.6.1 Examination of Product. A visual inspection of the compass shall be made to verify that the materials, design and construction, necessary mechanical measurements, marking and workmanship comply with the requirements of this specification.

4.6.2 Operating Speed. The time for the gyro motor to reach operating speed shall be noted and shall not exceed 2 minutes. Operating speed is defined as 40 percent of full speed.

4.6.3 Azimuth Servo Channel Operation. The compass shall be properly connected. The directional control shall be rapidly moved through an angle of approximately 45° about its azimuth axis. The indicator shall move smoothly without noticeable jumping or overshooting. There shall be no noticeable hunting or oscillation when the directional control is stationary or moving.

4.6.4 Follow-Up Rate. The compass shall be properly connected. The directional control shall be turned through an angle in excess of 360°, first clockwise and then counterclockwise, at a rate not less than 7 rpm. The indicator pointer shall rotate through 360° in 9 seconds or less for both clockwise and counterclockwise rotation.

*4.6.5 Normal Slaving Rate. The compass shall be set into operation in the magnetic mode and a precise indicator shall be electrically connected to the autopilot or indicator data transmitter. The compass shall be synchronized with the magnetic detector on north heading. The magnetic detector shall then be turned 20° clockwise and the time shall be recorded for the precise indicator to turn 12° clockwise. The time shall be between 6 and 12 minutes exclusive of the effect due to the earth's rotation.

4.6.5.1 The test of 4.6.5 shall be repeated with the magnetic detector turned 20° in a counterclockwise direction. The same results shall be noted except direction will be counterclockwise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.6 Individual Testing.

4.6.7 Random Wander.

4.6.7.1 For Individual Tests. The directional control shall be mounted in its normal operating position on a Scorsby table (or equivalent roll, pitch and yaw testing apparatus which shall provide 5 to 7 oscillations per minute) with the table in a level position and the remainder of the compass placed on a test bench. The compass shall be energized and the directional control permitted to reach operating speed with the spin axis approximately horizontal and on a random heading. Magnetic slaving and latitude correction shall be cut out for this test. The Scorsby table shall be made to oscillate in yaw, pitch and roll approximately 0.5° either side of the reference axis.

a. The extremes of the indications of the directional control shall be read and recorded for each 1/2-hour period for a total time not less than 8 hours.

b. The gyro drift for each hourly period shall be determined and shall not differ from the computed drift for the local latitude by more than $\pm 3°$.

c. The over-all drift of the gyro shall not differ from the computed drift for the same period by more than $\pm 3°$ per hour.

4.6.7.2 The power shall then be removed from the compass for a period not less than 4 hours and the tests of 4.6.7.1 repeated. The same tolerances shall apply.

4.6.7.3 For the purpose of this test, the directional control may be tested as an individual component.

4.6.8 For Sampling Tests. The test outlined under 4.6.7.1 shall be conducted except that the Scorsby table shall be made to oscillate in roll, pitch, and yaw approximately 7.5° either side of the reference axis and shall provide 5 to 5.5 oscillations per minute.

4.6.8.1 The power shall then be removed from the compass for a period not less than 4 hours and the tests of 4.6.8 repeated. The same tolerances shall apply.

4.6.8.2 The requirement of 4.6.7.3 is applicable.

4.6.9 Latitude Correction. The compass shall be operated with the latitude correction control set for periods not less than 1 hour each at 0° N, 30° N, 60° N, 90° N and 90° S. The compass shall be otherwise tested in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

with 4.6.7.1 The drift of the gyro for each hourly period shall not differ by more than ±4° per hour from the difference between the drift computed for the local latitude and the drift correction computed for the latitude of correction control setting.

*4.6.10 Balance. Leveling shall be cut out for this test. The directional control (nonoperating) shall be mounted on a tilt table so that the azimuth axis is perpendicular to a plane which is equivalent (parallel) to the tangent plane of the earth at the equator. The gyro spin axis shall be aligned North-South within five (5) degrees. The gyro shall be energized. When the gyro motor has reached synchronous speed, the motor shall be observed through the window as being case leveled.

*4.6.10.1 The drift shall be checked (under Scorsby) in the same manner as specified for the random wander test (4.6.7.1). The drift during a 1-hour period shall not vary more than 5.5° per hour from the computed drift. The computed drift shall equal 15° per hour times the sine of the local latitude.

4.6.10.2 This test may be accomplished before the directional control case is closed.

4.6.11 Gyro Card Stability. Magnetic slaving and leveling shall be cut out for this test. The gyro shall be brought up to speed. The directional control shall be mounted on a tilt table and tilted .55° ±1° from the vertical. The directional control shall then be rotated one complete revolution about a vertical axis at a speed of approximately 360° per minute. The drift of the gyro shall not exceed 2° as a result of this rotation.

4.6.11.1 The test of 4.6.11 shall be repeated, rotating the directional control in the opposite direction.

4.6.11.2 This test may be accomplished before the directional control case is closed.

4.6.11.3 For the purpose of this test, the directional control may be tested as an individual component.

4.6.12 Gimbal Freedom. Magnetic slaving and leveling shall be cut out for this test. The directional control, mounted on a turntable, without benefit of shockmounts, and operating at synchronous speed, shall be tilted 80° out of horizontal plane. The directional control shall then be rotated through 360° in azimuth. There shall be no indication of gimbal interference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.12.1 For the purpose of this test, the directional control may be tested as an individual component.

4.6.13 Spinning and Oscillation. Slaving shall be cut out for this test. The directional control shall be set into operation and the action of the gyro noted. There shall be no spinning, and any oscillation shall damp out in less than 30 seconds after application of power.

4.6.13.1 The power shall be removed from the compass. The gyro shall not oscillate or spin at a rate which would be detrimental to subsequent operation.

4.6.13.2 For the purpose of this test, the directional control may be tested as an individual component.

4.6.14 Leveling. Slaving shall be cut out for this test. The directional control shall be operated in its normal operating position. The directional control shall then be tilted in such a manner that the gyro axis shall be precessed approximately 20° out of level with the case. The leveling rate shall be determined to be between 2° and 4° per minute, as the gyro levels itself exclusive of the effect due to earth's rotation.

4.6.14.1 For gravity leveled gyros, this test may be accomplished before the directional case is closed.

4.6.14.2 For the purpose of this test, the directional control may be tested as an individual component.

4.6.15 Polarity.

4.6.15.1 Counterclockwise Direction. The compass shall be set into operation and manually synchronized on a 0-degree heading. The magnetic detector shall be turned 30° in a counterclockwise direction (numerically decreasing heading). The direction of rotation of the repeater indicator pointer shall be a numerically decreasing direction. The annunciator shall point to the operator's right in the sector designated by the minus sign. A counterclockwise rotation in the direction indicated for minus on the synchronizer control shall cause the annunciator to move to the left and synchronize with the magnetic detector when the annunciator reaches its vertical position.

4.6.15.2 Clockwise Direction. The procedure of 4.6.15.1 shall be repeated by turning the detector in a clockwise direction. The direction of rotation of the repeater indicator pointer shall be in clockwise direction and the annunciator shall point to the operator's left in the plus sector. A clockwise rotation in the direction indicated as plus on the synchronizer control shall cause the annunciator to move to the right and synchronize with the magnetic detector when the annunciator reaches its vertical position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.16 Annunciator. The compass shall be set into operation, slaved to the magnetic detector. The detector shall be mounted on a turntable suitable for rotating through an angle of "±0.25°." The compass shall be synchronized on magnetic north. The detector shall then be rotated through an angle of ±.25° about north. The annunciator shall indicate a perceptible right-left movement with respect to its synchronized vertical position.

4.6.17 Static Settling Test. The compass shall be set into operation and set to a synchronized null. The directional control shall then be rotated 10° from the null position. The output of the precise data transmitter shall be read after a period of 15 minutes has elapsed. This procedure shall be followed for both sides of null. The difference between the two final settling positions shall not be greater than ±0.25°.

4.6.18 Dynamic Accuracy. The compass shall be set into operation and set to a synchronized null. The magnetic detector shall be oscillated through ±5° either side of the null at 6hz per minute, and a 1-minute interval high and low reading shall be made. Readings shall continue until three consecutive 1-minute readings are within 0.1° of each other; this shall occur in less than 45 minutes from the start of the test. The difference between the initial null value and the average end value (determined by averaging the high and low readings of the last three consecutive readings) shall be no greater than 1.5°.

4.6.19 Scale Error.

4.6.19.1 A magnetic detector with known error shall be mounted on a suitable turntable in an area having magnetic quality which shall permit the performance of this test. The compass shall be connected and set into operation as a magnetic compass and synchronized on magnetic north. The detector shall be rotated in 15-degree increments through 360° clockwise and the compass manually set to a synchronized null on each heading. A unit containing a synchro of known calibration, mounted in an accurate positioning device and a null reading meter, or their equivalents, shall be electrically connected to the precise data transmitter for accurate determination of compass heading. These readings shall be recorded for each position of the magnetic detector.

4.6.19.2 The test of 4.6.19.1 shall be repeated for each successive 15-degree heading, from 0°, with a counterclockwise rotation of the magnetic detector. The deviations determined by both rotations shall be added algebraically and the result divided by the number of readings. This result shall be the index error and shall not exceed ±0.25°. The readings shall not differ from the correct heading, less the index error, by more than 0.5°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.19.3 The difference of like readings as a result of clockwise and counterclockwise rotation shall not exceed ±0.5°.

4.6.19.4 The electrical zero position for the indicator data transmitter shall be the same as that for the precise data transmitter within 0.28.

4.6.20 Turn Cutout Magnetic Slaved Operation.

*4.6.20.1 The compass shall be set into operation as a magnetic compass and shall be synchronized to the magnetic detector at 0 degree heading with the gyro leveled. The gyro spin axis shall be tilted 20°. The input to the one phase of the leveling torquer shall be noted. The turn cutout relay shall then be closed. The voltage to the same phase of the leveling torquer shall be less than 0.25 volt.

4.6.20.2 For gravity leveled gyros, this test may be accomplished before the directional control case is closed.

4.6.20.3. The magnetic detector shall be rotated 90° from the synchronized null. The differential current into the azimuth torquer shall be recorded. It shall be as required to obtain the proper slaving rate. The turn cut out relay shall be closed. The current to the windings of the azimuth torquer shall be less than 0.002 milliampere.

4.6.21 Synchronization.

4.6.21.1 Magnetic Mode.

4.6.21.2 Automatic Synchronization. The compass shall be set into operation as a magnetic compass and synchronized with the magnetic detector on a north heading. The electrical power shall then be removed, the magnetic detector rotated clockwise 170° and the electrical power re-applied to the compass. The compass shall synchronize to within 2° of the 170-degree heading in not more than 10 seconds from the start of motion of the indicator.

4.6.21.3 Manual Synchronization. At the conclusion of the automatic synchronization cycle, the magnetic detector shall be set on a north heading and the compass manually synchronized counterclockwise at the maximum rate. It shall take not more than 5 seconds to reach synchronization.

4.6.21.4 The electrical power shall then be removed, the magnetic detector rotated counterclockwise 170° and the electrical power re-applied to the compass. The test shall be repeated rotating the magnetic detector in the clockwise direction. The results shall be the same as specified in 4.6.21.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.21.5 DG Mode. After the compass has been automatically synchronized at the north headings, the procedures of 4.6.21.2 and 4.6.21.4 shall be repeated, except that the function selector switch shall be momentarily placed in the DG position and then returned to the MAG position in lieu of removing the electrical power from the compass. The compass shall synchronize to within 2° of the 170-degree heading in not more than 10 seconds from the start of motion of the indicators.

4.6.22 Autopilot Slewing Decoupling.

4.6.22.1 The compass shall be set into operation in the DG mode of operation. A Type V-7A, or comparable, indicator shall be connected to the autopilot output. There shall be no movement of the indicator dial when the slew control on the control panel is in operation.

4.6.22.2 The test of 4.6.22.1 shall be performed in the MAG mode of operation and the same results shall be noted.

4.6.23 Reliability Test. Each compass shall be subjected to a 5-hour reliability test. At least 99 percent (reference figure 5) of the equipments shall complete this test without failure, to comply with the mean-time-to-failure requirements of 3.5.6. The contractor shall submit a report at a 3-month interval indicating the following:

a. Number of equipments tested during the last 3-month period.

b. Number of equipments which failed.

c. Time in each case until failure.

d. Description of each failure.

4.6.24. Dielectric Strength and Insulation Resistance. Dielectric tests shall be made on all individual components between isolated pins and between pins and case, where no circuit is closed, for a period of 10 seconds. In all cases, the resistance shall be more than 10 megohms. The following direct current potentials shall be applied:

a. A potential of 280V dc between the case and pins common to the gyro motor of the direction control.

b. A potential of 500V dc between isolated pins and between pins and case on all components, except those pins common to the gyro mentioned in 4.6.24a.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

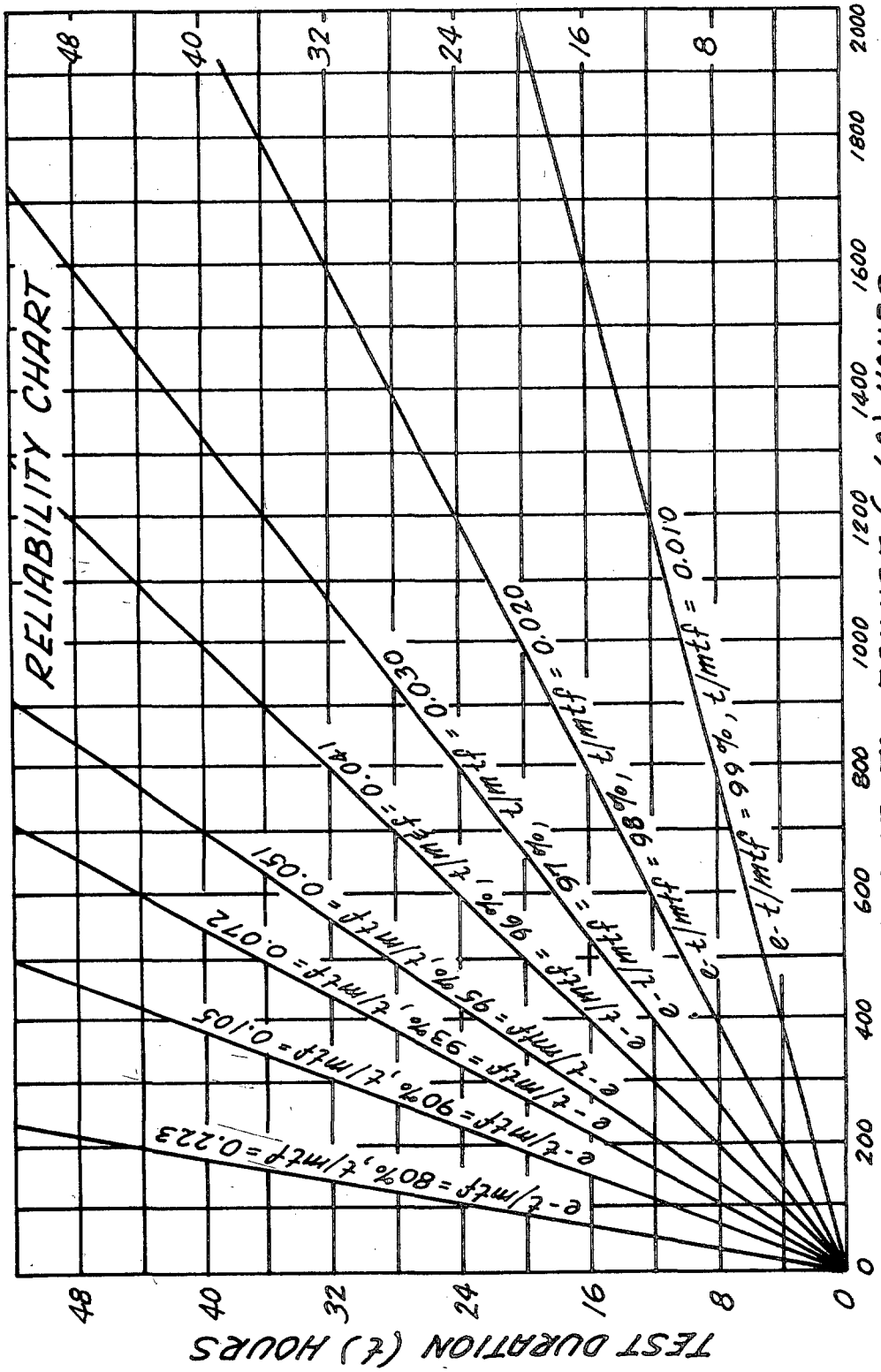


FIGURE 5

This chart, based on the exponential failure distribution, may be used as a criteria for determining when the equipment meets the mean-time-to-failure requirement. Several combinations of test conditions are available for each mean-time-to-failure requirement which may be placed on the equipment. For example, a mean-time-to-failure of 800 hours for a radar set requires 99 percent of the sets to operate without failure for a specified test duration of 8 hours or 97 percent for 24 hours, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.25 Low Temperature Operation. The compass shall be properly connected, without power, and subjected to a temperature of -54°C ±2°C (-65°F ±3°F) for a period of 4 hours. At the end of this period and while still at the low temperature, the compass shall meet the requirements specified herein for operating speed, azimuth servo channel operation, follow-up rate, normal slaving rate, random wander, latitude correction, gyro card stability, dynamic accuracy, and scale error, except as specified herein.

4.6.25.1 Operating Speed. The time required for the gyro motor to reach operating speed shall not exceed 6 minutes.

4.6.25.2 Random Wander. The drift of the gyro for each 1/2-hour period (through a 2-hour test) shall not exceed 2.25°.

4.6.25.3 Latitude Correction. The error shall not exceed ±5.5°per hour.

4.6.25.4 Gyro Card Stability. The drift of the gyro shall not exceed 2.5° as a result of the test specified in 4.8. This test shall run within 1 minute after removal from the test chamber.

4.6.25.5 Dynamic Accuracy. The error shall not exceed 2°.

4.6.25.6 Scale Error. The index error shall not exceed ±0.5°.

4.6.26 High Temperature Operation. The compass shall be properly connected, without power, and subjected to a temperature of 71°C ±2°C (160°F ±3°F) for a period of 4 hours. At the end of this period and while still at the the high temperature, the compass shall meet the requirements specified herein for operating speed, azimuth servo channel operation, follow-up rate, normal slaving rate, random wander, latitude correction, gyro card stability, dynamic accuracy and scale error, except as specified herein.

4.6.26.1 Operating Speed. The time required for the gyro motor to reach operating speed shall not exceed 4 minutes.

4.6.26.2 Random Wander. The drift of the gyro shall be as specified in 4.6.25.2.

4.6.26.3 Latitude Correction. The error shall be as specified in 4.6.25.3.

4.6.26.4 Gyro Card Stability. The drift of the gyro shall be as specified in 4.6.25.4.

4.6.26.5 Dynamic Accuracy. The error shall be as specified in 4.6.25.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.26.6 Scale Error. The index error shall be as specified in 4.6.25.6.

4.6.27 Power Consumption. The power consumption of the compass shall be checked under the following starting and normal operating conditions:

a. The ac power and dc power consumption shall not exceed 80 volt-amperes and 15 watts, respectively, during starting operation.

b. The ac power and dc power consumption shall not exceed 75 volt-amperes and 15 watts, respectively, for normal operation.

4.6.27.1 The heater power shall be intermittent and shall not exceed an additional 30 watts dc.

4.6.28 Voltage and Frequency Variation.

4.6.28.1 The compass shall be set into operation and synchronized on a 0-degree heading. The voltage and frequency of the ac supply shall be lowered to 104 volts, 360hz and the dc supply lowered to 25 volts. The shift in the synchronized null as read on a precise indicator connected to the amplifier precise data transmitter shall not exceed 0.75°. At these low voltages and frequency the compass shall be resubjected to the random wander (4.6.8) and latitude correction tests (4.6.9) and an additional tolerance of 10 percent shall apply.

4.6.28.2 The voltage and frequency of the ac supply shall then be raised to 126 volts, 440hz and the dc supply raised to 30 volts. The test procedure of 4.6.28.1 shall be repeated and the same tolerances shall apply.

4.6.29 Low Field Strength. The compass shall be set into operation on a synchronized null and then subjected to the dynamic accuracy test (4.6.18) except that the oscillating signal shall be equivalent to a magnetic detector oscillating about the reference null setting in a horizontal magnetic field of 0.080, +0.002, -0.000 oersted and in a vertical field of 0.880 $\pm$ 0.002 oersted. The resulting shift in synchronized null shall not exceed 2°. The test shall be run on three headings not less than 60° apart.

4.6.30 Rotational Test. The directional control shall be mounted on a fixture which shall permit 360° of rotation in a horizontal axis, perpendicular to the azimuth axis.

a. The case shall be placed in the proper operating position, the gyro brought up to speed, and with the unit at the normal operating level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

(1) Start with a 0° ±2° azimuth heading of the gyro,

(2) Rotate the test fixture about a horizontal axis, by hand, clockwise and counterclockwise approximately 65°,

(3) Observe on a precise data indicator the movement of the inner gimbal relative to the case.

b. The gyro assembly may be moved on the mounting plate in azimuth until a minimum movement is discernible on the indicator for each direction of rotation.

(1) Firmly lock the mounting plate,

(2) Torque the gyro to a 0° ±2° heading,

(3) Recheck the results.

c. If the minimum displacement of the indicator dial is observed under the conditions of 4.6.30b proceed with the test; otherwise, repeat as outlined until the proper alignment is obtained.

(1) Note this heading,

(2) Then rotate the fixture at approximately 20 rpm for 360° causing the azimuth gimbal of the gyro to go through a double reversal while the fixture is completing one revolution back to its original position.

d. The heading of the gyro shall again be noted and the resulting difference in azimuth reading shall not exceed 7°.

(1) Return the gyro to the 0° ±2° heading and the normal operating level,

(2) Then repeat this test, rotating the unit through 360° at 50 rpm in both a clockwise and counterclockwise direction, with the level of the spin axis not more than 60° from horizontal at the conclusion of one complete revolution in either direction.

4.6.30.1 No physical damage shall result from this test.

4.6.31 Vibration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

4.6.31.1 Directional Control Vibration. With the directional control energized and the spin axis approximately horizontal and at any random heading, the directional control shall be subjected to a vibration test, procedure V of MIL-E-5272. At the end of this test the directional control shall meet the specified requirements for random wander (4.6.8).

4.6.31.2 For the purpose of this test, the directional control may be tested as an individual component.

4.6.31.3 Servo-Amplifier Vibration. The servo-amplifier shall be subjected to a vibration test, procedure V of MIL-E-5272. There shall be no variation in the outputs of the data transmitters during this test.

4.6.31.4 Control Panel Vibration. The control panel shall be subjected to a vibration test, procedure IV of MIL-E-5272, with excursions of 0.002 to 0.020 inch. There shall be no movement of any of the controls and the annunciator needle shall not vibrate excessively during this test. There shall be no damage to the control panel as a result of this test.

*4.6.32 Radio Noise Interference. Radio noise measurements shall be made in a shielded room and under the following conditions:

a. Conducted Noise. Conducted radio noise measurements shall be made in accordance with MIL-STD-461.

b. Radiated Noise. Radiated radio noise measurements shall be made on the equipment in the range from 0.15 to 20 megahz in accordance with MIL-STD-461.

c. Measurements.

(1) Radiated radio noise measurements shall be made on the compass in the range from 20 to 150 megahz using a Measurements Corporation Model 58, Very High Frequency Noise Meter with dipole antenna, or approved equivalent. The arrangement of compass and test apparatus shall be as specified in MIL-STD-461.

(2) When making any measurements, the "Peak-Average" switch of the Model 58 instrument shall be in PEAK position. Measurements shall be made on not less than 12 frequencies within the required range and not less than three measurements shall be made in each of the frequency bands A, B, C and D of the test instrument. Measurements shall be made also at any points of peak noise output found in sweeping the specified frequency range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

d. The compass shall be operated in both of the two standard modes of operation, MAG and DG, during testing. An additional tolerance of 16 db above the transient limits shall apply to tests run during turn cut out operation.

4.6.33 Environmental Tests.

*4.6.33.1 Sampling procedure shall be as specified in paragraph 4.4.3 Sampling Plan and Tests, and paragraph 4.4.2 Individual Tests. Each component of the compass shall be subjected to the following tests in accordance with MIL-E-5272:

a. High Temperature Test, Procedure I. The test shall be for a period of 24 hours. The compass components shall be properly connected, without power. At the end of this period and while still at the high temperature, the azimuth servo channel, as indicated by a Type V-7A repeater indicator at room temperature, or other means acceptable to the procuring activity, shall follow the movement of the directional control without binding or sticking. After the compass has returned to room temperature, it shall meet all individual tests specified.

b. Humidity, Procedure I. At the conclusion of the humidity test there shall be no evidence of corrosion of the internal or external parts of the individual components of the compass. The compass shall meet the specified requirements for manual synchronization and normal slaving rates.

* c. Salt Spray, Procedure I. The test shall be for a period of 50 hours. External connections shall be made to the compass in such a manner as to simulate installed conditions. At the end of the 50-hour period, the compass shall be subjected to the individual tests specified herein without recourse to brushing or washing and the same tolerances shall apply. At the end of the 50-hour period, the compass shall be examined per FED-STD-151, Method 811. The compass shall then be subjected to the individual tests specified herein.

d. Vibration, Procedure V. After these vibration tests, the compass shall meet the requirements specified herein for random wander (4.6.8), latitude correction rate, dynamic accuracy and scale error. No looseness in the mechanism or damage to any part of the compass components shall result from the vibration.

e. Rain, Procedure I. At the conclusion of this test, there shall be no evidence of water penetration or damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

f. Sand and Dust, Procedure I. At the conclusion of this test, the equipment shall be examined and operated. There shall be no evidence of sand or dust penetration or damage.

4.6.33.2 High Altitude-Low Temperature Test. The component shall be subjected to a pressure equivalent to an altitude of 70,000 ±500 feet at a temperature of -54°C ±3°C (-65°F ±5°F) for a continuous period of 4 hours. While at this pressure and temperature, the component shall be subjected to the low temperature operation (4.6.25) with exception of the gyro card stability requirement, and the same tolerances shall apply. After the component has returned to room temperature, it shall meet all individual tests.

*4.7 Inspection and Test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this Specification.

*5 PREPARATION FOR DELIVERY.

5.1 Preservation and Packaging. Detailed packaging requirements for components of Compass, Magnetic, Slaved Gyro, Type J-4 shall be as stipulated in the following paragraphs.

5.1.1 Directional Control to be preserved and packaged in Quality Unit Packs of one (1) each in accordance with Method IIB of MIL-P-116. Detailed requirements for wraps, special dunnage and reusable containers shall comply with Transportation Packaging Order (TPO) 6615-515-6000. Copies of referenced TPO may be obtained from the Prime IM/AMA/SPC office.

5.1.2 Servo Amplifier to be packaged in quantity unit packs of one (1) each preserved method IIB in accordance with MIL-P-116. Item to be wrapped with a polyfilm, L-P-378 or equal. Fiberboard dunnage cells, sleeves, pads, die cuts, etc, constructed in accordance with PPP-F-320 to be utilized for the purpose of immobilization of the amplifier within the primary container. Apply two (2) inches of boundfiber MIL-C-7769 type III, class A to all faces between primary and unit container. Primary and unit container to be constructed in accordance with PPP-B-636, Stype RSC.

5.1.3 Control panel to be packaged in quantity unit packs of one (1) each and preserved a method IIC in accordance with MIL-P-116. Wrap panel in L-P-378 polyfilm or equal. Cushion all edges, knobs, and dials with PPP-C-843 cellulosic cushioning material to protect item and prevent bag puncture. Apply two (2) inches of MIL-C-7769 type III cushioning material to all faces of pack between bagged item and unit container. Unit container to be constructed in accordance with PPP-B-636, Stype RSC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

5.2 Packing. Each directional control, servo amplifier or control panel procured in accordance with this specification shall be packed Level "B" unless otherwise stipulated by the procuring activity.

*6 NOTES.

6.1 Intended Use. The Type J-4 compass covered by this specification is intended for use in fighter, short range bomber and cargo type aircraft whose range of operation includes the polar regions as well as all aircraft requiring medium precise heading reference stabilization.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see Section 5 of this specification).

6.3 First Article Inspection. It is expected that the contract or purchase order will specify that three compasses will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.4 Definitions.

6.4.1 Amplitude. The work "amplitude", as specified herein, shall mean 1/2 of the total excursion.

6.4.2 Apparent Drift. Apparent drift is the apparent motion of the directional control inner gimbal with respect to the earth. This motion is due to the rotation of the earth.

6.4.3 Synchronized Null. "Synchronized Null" is the condition when the compass is completely synchronized with the DT-309/AJN detector, or electrical equivalent.

6.4.4 International Standardization. Certain provisions (see 3.5.2.1, 3.7.1.2, 3.7.1.3.2, 3.7.2.1, 3.7.2.2.1, 3.7.2.2.2, 4.5.5, 4.6.27, 4.6.28) of this specification are the subject of international standardization agreement (ASCCAS 53/10). When amendment, revision, or cancellation of this specification is proposed, which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assures no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force 71

Preparing Activity:
Air Force 71

Project Number
6605-F164

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:24Z
Check the source to verify that this is the current version before use.